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Studierichting: Informatica
Oefeningenreeks 6B nr. 4.5

$$x_1 \text{ en } v \text{ als } s_7 = -42 \text{ en } s_{10} = -120$$

$$s_7 = \frac{x_1 + x_7}{2} \cdot 7 = -42$$

$$= \frac{x_1 + (x_1 + 6v)}{2} \cdot 7 = -42$$

$$= \frac{2x_1 + 6v}{2} \cdot 7 = -42$$

$$= (x_1 + 3v) \cdot 7 = -42$$

$$x_1 + 3v = -6$$

$$s_{10} = \frac{x_1 + x_{10}}{2} \cdot 10 = -120$$

$$= \frac{x_1 + (x_1 + 9v)}{2} \cdot 10 = -120$$

$$= \frac{2x_1 + 9v}{2} \cdot 10 = -120$$

$$= \left(x_1 + \frac{9}{2}v\right) \cdot 10 = -120$$

$$x_1 + \frac{9}{2}v = -12$$

$$\Rightarrow (x_1 + 3v) - \left(x_1 + \frac{9}{2}v\right) = -6 - (-12)$$

$$-\frac{3}{2}v = 6$$

$$v = -4$$

$$\Rightarrow x_1 + 3(-4) = -6$$

$$x_1 - 12 = -6$$

$$x_1 = 6$$

References